A Post-Quantum Compliant Authentication Scheme for IoT Healthcare Systems

Morteza Adeli, Nasour Bagheri, Hamid Reza Maimani, Saru Kumari, *Member, IEEE*, and Joel J. P. C. Rodrigues, *Fellow, IEEE*

***Abstract*—In an Internet of Things (IoT)-based healthcare system, medical IoT devices gather and transmit critical patient data. Ensuring the security and privacy of medical data is paramount. One of the most critical challenges in this regard is the authentication of participating entities. The literature proposes specific authentication approaches for healthcare systems based on integer factorization and discrete logarithm problems. However, the advent of quantum computers would fundamentally break all of these protocols. In this study, we conducted an analysis of a recently proposed authentication and access control scheme for e-health systems, which is based on lattice-based cryptography and was developed by Gupta et al. Our analysis revealed that the scheme is vulnerable to several types of attacks, including impersonation, de-synchronization, and smart card stolen attacks, which could compromise the confidentiality and integrity of sensitive medical data. To address these security challenges, we propose an alternative authentication and access control scheme that uses *Saber*, a finalist lattice-based key encapsulation algorithm from round three of the NIST post-quantum cryptography standardization. One of the biggest advantages of Saber is its simplicity and efficiency. Our proposed scheme is designed specifically for e-health systems and provides robust protection against the vulnerabilities identified in Gupta et al.'s scheme. We believe that our proposed scheme represents a significant improvement over existing approaches and could help to enhance the security and privacy of e-health systems. Upon completion of our improved protocol, we proceeded to implement it within the Vivado 2018.3 environment for Zynq UltraScale FPGAs. To gather insight into its performance, we conducted a performance comparison study with various related protocols.**

***Index Terms*—Authentication, FPGA, Internet of Things (IoT), lattice-based cryptography, security analysis.**

I. Introduction

THE GROWTH of the Internet of Things (IoT) has brought about a network of interconnected smart devices that includes radio frequency identification (RFIDs), sensors, desktop computers, mobile phones, and more. One significant application of IoT is in healthcare, where it has the potential to improve patient's quality of life by facilitating faster access to better treatments. However, securing personal information collected, stored, or transmitted by medical IoT devices remains a significant challenge in healthcare IoT systems, as depicted in Fig. 1. Ensuring the security, integrity, and confidentiality of such information is paramount. Authentication of users is a critical challenge for IoT healthcare systems. To address this challenge, various authentication schemes leveraging different types of cryptographic primitives like hash functions, elliptic curve functions, and others have been proposed in [1] and [2]. Currently, most public key authentication schemes rely on two problems: 1) integer factorization (e.g., RSA) [3] and 2) discrete logarithms (e.g., ECC) [4]. Unfortunately, quantum computers can break these algorithms within a short time [5]. In response to this vulnerability, post-quantum cryptography (PQC) algorithms have been developed. These algorithms offer protection against both classical and quantum attacks and are designed to work compatible with existing communication protocols. There are five primary classes of PQC algorithms: 1) lattice-based cryptography [6]; 2) code-based cryptography [7]; 3) multivariate-based cryptography [8]; 4) hash-based cryptography [9]; and 5) isogeny-based cryptography [10]. Lattice-based cryptography is considered a top contender for future PQC standard algorithms. In 2022, NIST announced that the lattice-based algorithm CRYSTALS-KYBER has entered to 4nd round of NIST PQC standardization and recommended it for implementation in most use cases. This is because many of the lattice-based cryptosystems are secure and involve simple and parallelizable computations that can be executed with great efficiency using conventional computing methods. In recent years, there has been a significant focus on PQC, particularly in the context of authentication protocols [11], [12]. However, to make optimal use of PQC, current authentication protocols based on the discrete logarithm or integer factorization problem will most likely need to be redesigned. In recent years, some PQC authentication protocols for IoT environments have been proposed by researchers [13], [14]. However, they have certain security vulnerabilities, weaknesses, and limitations [15]. Recently, Gupta et al. [16] proposed a lattice-based authentication and access control (LAAC) scheme specifically tailored for healthcare systems. They claimed that their scheme

Manuscript received 13 May 2023; revised 28 July 2023; accepted 23 August 2023. Date of publication 5 September 2023; date of current version 6 February 2024. This work was supported by the Shahid Rajaee Teacher Training University under Grant 4968. *(Corresponding authors: Nasour Bagheri; Saru Kumari.)*

Morteza Adeli is with the Faculty of Electrical and Computer Engineering, Malek Ashtar University of Technology, Tehran 1774-15875, Iran (e-mail: m.adeli@sru.ac.ir).

Nasour Bagheri is with the Department of Electrical Engineering, Shahid Rajaee Teacher Training University, Tehran 1678815811, Iran, and also with the School of Computer Science, Institute for Research in Fundamental Sciences (IPM), Tehran 19538-33511, Iran (e-mail: Nbagheri@sru.ac.ir).

Hamid Reza Maimani is with the Department of Science, Shahid Rajaee Teacher Training University, Tehran 1481867618, Iran (e-mail: maimani@ipm.ir).

Saru Kumari is with the Department of Mathematics, Chaudhary Charan Singh University, Meerut 250004, India (e-mail: Saryusiirohi@gmail.com).

Joel J. P. C. Rodrigues is with the COPELABS, Lusófona University, 1749-024 Lisbon, Portugal (e-mail: joeljr@ieee.org).

Digital Object Identifier 10.1109/JIOT.2023.3309931

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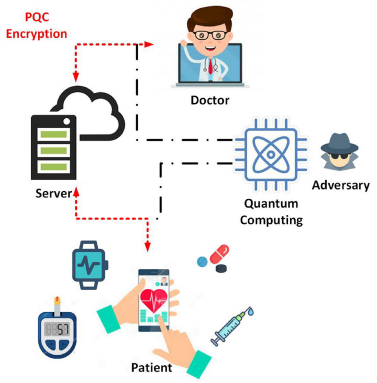


Fig. 1. IoT healthcare system.

TABLE I
NOTATION USED IN THIS ARTICLE

Notation	Description
$Ul_i/S/D_j$	User/Server/Device
$H_j(.), j = 1, 2, 3$	Hash functions
ID_i, PWD_i	Identity, password of U_i
ID_{i_t}	Temporary identity of U_i
T_i	Timestamp
$()_L, ()_R$	Left bits,Right bits
$d \in \mathbb{Z}_q^n$	Master secret
$PU \in \mathbb{Z}_q^{1\times m}$	Public key
Saber.PKE.Enc	Saber public key encryption
Saber.PKE.Dec	Saber public key decryption

is secure against both quantum and common classical known attacks. However, in this article, we conduct a more detailed analysis of the security of this protocol and reveal its vulnerability to certain attacks. Throughout this article, we use the notations that are listed in Table I.

A. Our Contribution

This article has two main contributions.

1) In this analysis, we examine the security of an authentication scheme proposed by Gupta et al. [16], designed for use in healthcare systems. The authors employ lattice-based cryptography to ensure the scheme's resistance to quantum computation attacks. However, we have identified vulnerabilities in their proposal, including susceptibility to impersonation, de-synchronization, and smart-card stolen attacks.
2) We have proposed an improved authentication scheme to address the vulnerabilities we identified in Gupta et al.'s original proposal. Our revised scheme is based on a lattice-based algorithm from the third round of the NIST PQC standardization process. We implemented our improved scheme using Verilog language on Zynq UltraScale FPGAs. Additionally, we conducted both formal and informal security analyses to investigate the effectiveness of our proposed solution.

B. Organization

The remainder of this article is organized as follows: Section II provides an introduction to the basic concepts and preliminaries of lattice-based cryptography. We describe the LAAC scheme in Section III. In Section IV, we identify security vulnerabilities present in the LAAC scheme proposed by Gupta et al. Our improved authentication scheme is presented in Section V. Next, we analyze the security of our improved scheme in Section VI and provide a performance comparison with other authentication schemes in Section VII. Finally, we conclude our work in Section VIII.

II. PRELIMINARIES

A. Lattice-Based Cryptography

A lattice $\mathcal{L}$ is the set of all integer linear combinations $\{\sum b_i x_i | x_i \in \mathbb{Z}\}$ of basis vectors $b_1, b_2, \ldots, b_n \in \mathbb{R}^m$. We define an $m \times n$ matrix $\mathcal{B}$ whose columns are $b_1, b_2, \ldots, b_n$ as a generator matrix for the lattice $\mathcal{L}$. The values n and m represent the rank and the dimension of the lattice $\mathcal{L}$, respectively. The lattice $\mathcal{L}$ is the full rank when $n = m$. A lattice $\mathcal{L}$ has infinitely many different bases, but any two bases are related by a unimodular transformation. All lattice-based algorithms (PKE/KEM/KEX) proposed in the NIST PQC standardization process are based on the two problems *n*th order truncated polynomial ring unit (NTRU) [17] and *the learning with errors* (LWE) [18]. Generally, in the LWE problem, we try to guess the linear n-array function f using samples $y_i = f_i(x)$ some of them are perturbed by adding a small amount of noise.

The precise definition of the LWE problem is as follows.

Definition (LWE Problem [18]): Let λ be the security parameter, $n = n(\lambda)$, $m = m(\lambda)$, $q = q(\lambda)$ be integers and let $\chi = \chi(\lambda)$ be a distribution over $\mathbb{Z}_q$. The $LWE_{n,m,q,\chi}$ assumption says that, if we choose $A \xleftarrow{\$} \mathbb{Z}_q^{m\times n}$, $s \xleftarrow{\$} \mathbb{Z}_q^n$, $e \leftarrow \chi^m$, $u \xleftarrow{\$} \mathbb{Z}_q^m$ then the following distributions are computationally indistinguishable:

$$(A, A.s + e) \approx (A, u).$$

There is another approach called *the learning with rounding (LWR)* problem [19] to hide the low-order bits of a vector equation in the above definition. The idea of this approach is simple and straightforward, we simply round a vector equation instead of perturbing it by adding a small error. Let $\lfloor . \rfloor_p : \mathbb{Z}_q \to \mathbb{Z}_p$, where $q \geq p \geq 2$ is a rounding function

$$\lfloor x \rfloor_p = \lfloor (p/q).\bar{x} \rfloor \bmod p$$

where $\bar{x} \in \mathbb{Z}$ is any integer congruent to $x \bmod q$. When using LWE-based encryption schemes, there are two methods for agreeing on an exact (as opposed to approximate) shared key: 1) the *encryption-based* approach [20] and 2) the *reconciliation-based* approach [21]. The reduction of bandwidth requirement is the main advantage of the reconciliation-based approach.

Saber is among the finalists in round three of NIST PQC standardization and is based on the module LWR problem (Mod-LWR). The algorithm creators claim that it is an efficient, simple, and flexible solution that can replace current public key algorithms. There are several reasons to support these claims. First, by using powers of 2 as the integer module, there is no need for modular reduction or rejection sampling. Additionally, Saber offers the advantage of requiring only half the amount of randomness as compared to LWE. This results in a reduction of bandwidth overhead. Moreover, Saber is extremely flexible and allows for various security levels using just one core component.

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B. Adversary Threat Model

Through the security analysis of a protocol, we make the following assumptions:

1) The communication channel between the user and the server is public. Therefore, an adversary $\mathcal{A}$ can eavesdrop, modify, replay, discard, or block the messages.
2) Adversary $\mathcal{A}$ can use side-channel analysis attacks to exploit the information stored on the smart cards.
3) Adversary $\mathcal{A}$ has quantum computing power. Therefore, they can break classic public key algorithms, such as RSA and Elgamal. The symmetric encryption algorithms with a large enough key size are safe.
4) Based on Zipf's law [22], in the real applications, $\|D_{ID_i}\| \leq \|D_{PWD_i}\| \leq 10^6$ where D_{ID_i}, D_{PWD_i} are space of identities and space of passwords, respectively. To safeguard against an exhaustive search attack on the identity and password, we implement preventive measures to restrict $\mathcal{A}$ from carrying out such actions.
5) The server is protected, so an adversary cannot access the security data stored on it.
6) The communication channel between the server and the patient is secure.

III. LAAC Scheme

This section provides a concise overview of the LAAC scheme [16].

A. Set Up Phase

This phase is executed by server $\mathcal{S}$ as following.

1) It chooses a prime number p, a matrix $X \in \mathbb{Z}_q^{m\times n}$, two hash functions $H_1 : \{0,1\}^* \rightarrow \{0,1\}^k$ and $H_2 : \mathbb{Z}_q^n \times \{0,1\}^* \rightarrow \{0,1\}^k$.
2) $\mathcal{S}$ chooses uniformly random its own secret key $d \in \mathbb{Z}_q^n$ and computes public key $PU = d^T.X^T \in \mathbb{Z}_q^{1\times m}$.
3) At the end, $\mathcal{S}$ saves d as a secret and broadcasts the values $\Delta = \{m, n, q, X, PU, H_1(\cdot), H_2(\cdot)\}$.

B. Registration Phase

This phase uses a secure channel to transfer the registration parameters. When this phase is completed, a smart card is issued to a user $\mathcal{U}_i$.

1) User $\mathcal{U}_i$ chooses two parameters ID_i and PWD_i.
2) It chooses a random $r_i \in \mathbb{Z}_q$ and computes $\beta_i = H_1(r_i\|ID_i\|PWD_i)$. Then, it sends the values $\{ID_i, \beta_i\}$ to the server $\mathcal{S}$.
3) $\mathcal{S}$ chooses a secret $a \in \mathbb{Z}_q$ and computes $\gamma_i = H_2(d\|ID_i\|a)$, $\lambda_i = \beta_i + \gamma_i$. Then the parameters $\{m, n, q, X, PU, H_1(\cdot), H_2(\cdot), a, \lambda_i\}$ are stored on a smart card and delivered to the user $\mathcal{U}_i$. In this system, a is a common secret for all users.
4) $\mathcal{U}_i$ uses their smart card to compute $\gamma_i = \lambda_i \oplus H_1(r_i\|ID_i\|PWD_i)$, $\eta_i = H_1(ID_i\|PWD_i) \oplus r_i$ and $\zeta_i = H_1(H_1(ID_i\|PWD_i)\|r_i\|\gamma_i)$. Then he stores $\{m, n, q, a, X, PU, H_1(\cdot), H_2(\cdot), \lambda_i, \eta_i, \zeta_i\}$ on their smart card.

C. Login Phase

The login process of LAAC operates in the following manner:

1) $\mathcal{U}_i$ inserts their smart card into the reader and enters their identity ID_i along with their password PWD_i. The smart card calculates $r_i' = H_1(ID_i\|PWD_i)\oplus\eta_i$, $\gamma_i' = \lambda_i \oplus H_1(r_i'\|ID_i\|PWD_I)$ and $\zeta_i' = H_1(H_1(ID_i\|PWD_i)\|r_i'\|\gamma_i')$. Then it checks whether $\zeta_i \stackrel{?}{=} \zeta_i'$ is true or false. If the statement is false, the reader terminates the session. However, if it is true, the reader authenticates the user $\mathcal{U}_i$ and jumps to the next step.
2) The smart card performs the following computations using a timestamp T_1 and a random vector $u_i \in \mathbb{Z}_q^m$ as inputs. $t_i = u_i^T \cdot X \in \mathbb{Z}_q^{1\times n}$, $v_i = PU \cdot u_i$, $\delta_i = H_1(T_1\|ID_i\|\gamma_i')$, $\phi_i = H_1(v_i) \oplus (ID_i\|\delta_i)$. Then it sends $\{t_i, \phi_i, T_1\}$ to server $\mathcal{S}$ over a public channel.

D. Verification Phase

The verification phase of the protocol is executed as follows:

1) $\mathcal{S}$ checks $|T_2 - T_1| \leq \Delta T$. Then it computes $v_i' = d^T \cdot t_i^T$ and extracts ID_i and δ_i from $(ID_i\|\delta_i) = H_1(v_i')\oplus\phi_i$. Now it performs a search and checks whether the ID_i exists within its database. If it finds a match, it computes $\gamma_i = H_2(d\|ID_i\|a)$ and $\delta_i' = H_1(T_1\|ID_i\|\gamma_i)$. If $\delta_i \stackrel{?}{=} \delta_i'$ then $\mathcal{S}$ authenticates the user $\mathcal{U}_i$ and takes a fresh timestamp T' and sends it to them.
2) $\mathcal{U}_i$ extracts their timestamp T'' and checks $|T'' - T'| \leq \Delta T$. If the statement is determined to be true by $\mathcal{U}_i$, it will send T' back to the server. Otherwise, the session will be rejected.

E. Access Control Phase

During the access control phase, the following steps are performed:

1) Upon receiving T' at time T''', $\mathcal{S}$ checks $|T''' - T'| \leq \Delta T'$. Then it chooses a random integer $b_i \in \mathbb{Z}_q$ and computes $\theta_i = H_1(ID_{i_t}\|b_i)$, where ID_{i_t} is the temporary identity of $\mathcal{U}_i$ and is valid only in this phase. $\mathcal{S}$ computes a response message $R_i = (ID_i\|\theta_i\|ID_{i_t}\|T_3) \oplus H_1(v_i')$ and sends it to $\mathcal{U}_i$. T_3 represents the allocated time for this phase. At the same time, S also sends $\{ID_{i_t}, T_3, b_i\}$ to device D_j through a secure channel.
2) $\mathcal{U}_i$ calculates $(ID_i\|\theta_i\|ID_{i_t}\|T_3) = R_i \oplus H_1(v_i')$ and then verifies whether ID_i is a valid identity. If this condition is satisfied, $\mathcal{U}_i$ authenticates server $\mathcal{S}$. Then user $\mathcal{U}_i$ temporarily saved the values θ_i and ID_{i_t} in its smart card until this phase is completed. Next, $\mathcal{U}_i$ inputs the values ID_i, PWD_i into his/her smart card.
3) The smart card computes $r_i' = H_1(ID_i\|PWD_i) \oplus \eta_i$, $\gamma_i' = \lambda_i \oplus H_1(r_i'\|ID_i\|PWD_i)$ and $\zeta_i' = H_1(H_1(ID_i\|PWD_i)\|r_i'\|\gamma_i')$. Then it checks $\zeta_i \stackrel{?}{=} \zeta_i'$. If it is true, it computes $\pi_i = H_1(\theta_i) \oplus H_1(ID_{i_t}\|a\|T_4)$ and sends $\{\pi_i, ID_{i_t}, T_4\}$ to the device D_j where T_4 is current timestamp of $\mathcal{U}_i$.
4) The device D_j checks $|T_5 - T_4| \leq \Delta T$ where T_5 is current timestamp of D_j. Then it calculates $\pi_i' = H_1(H_1(ID_{i_t}\|b_i)) \oplus H_1(ID_{i_t}\|a\|T_4)$ and checks whether

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$\pi_i \stackrel{?}{=} \pi_i'$. If it is true, it authenticates user $\mathcal{U}_i$ and computes the session key $SK = H_1(T_3 \oplus \theta_i \oplus a)$. It encrypts data $Data_j$ as $C_{ij} = Data_j \oplus SK$ and sends it to user $\mathcal{U}_i$.

5) $\mathcal{U}_i$ decrypts C_{ij} as $Data_j = C_{ij} \oplus SK$.

IV. Security Challenges of the LAAC Scheme

1) *Impersonation Attack:* We know that the two values $(\phi_i)_L$ and $(R_i)_L$ are equal to $(h(v_i))_L \oplus ID_i$. The right bits $(\phi_i)_R = (h(v_i))_R \oplus \delta_i$ step 2 of the login phase that is performed between a user $\mathcal{U}_i$ and a legitimate server $\mathcal{S}$, $\mathcal{U}_i$ generates the values $\{t_i, \phi_i, T_1\}$ and sends it to server $\mathcal{S}$ and in step 1 of access control phase, server $\mathcal{S}$ sends back the value $\{R_i\}$ to $\mathcal{U}_i$. Therefore, if an adversary $\mathcal{A}$ eavesdrops on the communication between server $\mathcal{S}$ and user $\mathcal{U}_i$ in a valid session of the protocol, they will obtain knowledge of the value of $h(v_i) \oplus ID_i$. Based on this fact, let an adversary $\mathcal{A}$ have eavesdropped on the last communication between $\mathcal{U}_i$ and $\mathcal{S}$ of a valid session and knows the value $h(v_i) \oplus ID_i$. Therefore, they can impersonate a trusted server as follows.
   a) Upon receiving $\{t_i, \phi_i, T_1\}$, $\mathcal{A}$ impersonates $\mathcal{S}$ by selecting T' and sending it to $\mathcal{U}_i$.
   b) $\mathcal{U}_i$ checks whether the condition $|T' - T''| \leq \Delta T$ is true or false. If the condition is determined to be true, then $\mathcal{U}_i$ sends back T' to the adversary $\mathcal{A}$.
   c) $\mathcal{A}$ selects $b_i \in \mathbb{Z}_q, ID_{i_t}$ and computes $R_i' = (\phi_i)_L \| ((R_i)_R \oplus \Delta T_3)$.
   d) $\mathcal{U}_i$ computes $ID_i = (R_i')_L \oplus (h(v_i))_L$ and authenticates the server $\mathcal{S}$.
2) *De-Synchronization Attack:* In the proposed scheme, user $\mathcal{U}_i$ and device $\mathcal{D}_j$ use the permitted time for access phase T_3 to produce the shared session key SK. In step 1 of the access control phase, an adversary $\mathcal{A}$ changes the message $R_i = (ID_i \| \theta_i \| ID_{i_t} \| T_3) \oplus h(v_i')$ to $R_i' = (ID_i \| \theta_i \| ID_{i_t} \| T_3') \oplus h(v_i')$ and sends it to the user $\mathcal{U}_i$. At the same time, server $\mathcal{S}$ sends message $\{ID_{i_t}, T_3, b_i\}$ to the device $\mathcal{D}_j$ over a secure channel. Therefore, $\mathcal{U}_i$ and $\mathcal{D}_j$ produced two different session keys $SK' = (T_3' \oplus \theta_i \oplus a)$ and $SK = (T_3 \oplus \theta_i \oplus a)$, respectively.
3) *Smart-Card Stolen Attack:* Suppose that the smart card of an user $\mathcal{U}_i$ is stolen by an adversary $\mathcal{A}$. Consequently, they can extract the secret information stored in it by physical attacks, such as side-channel analysis. $\mathcal{A}$ knows the common value a, so they can compute the session key SK as following.
   a) $\mathcal{A}$ impersonates itself as a legitimate server. They compute $R_i' = (ID_i \| \theta_i \| ID_{i_t} \| a) \oplus H_1(v_i')$ and send it to the $\mathcal{U}_i$. Adversary $\mathcal{A}$ also can compute $h(\theta_i) = \pi_i \oplus h(ID_{i_t} \| a \| T_4)$ because they know π_i, ID_{i_t}, T_4.
   b) $\mathcal{U}_i$ computes $SK = h(a \oplus \theta_i \oplus a) = h(\theta_i)$.

V. SKEBA Scheme

This section aims to address the vulnerabilities of the LAAC scheme by introducing an improved lattice-based authentication scheme that is specifically designed for healthcare IoT environments. The primary challenge associated with lattice-based algorithms lies in the size of ciphertext and public keys. The next challenge is to design and implement an authentication protocol for resource-limited systems. The main rationale behind our proposed scheme is twofold:

1) minimizing the number of transferred messages between the parties;
2) reducing computation overhead.

We investigate lattice-based candidate algorithms in the 3rd round of the NIST PQC standardization process. Saber [23] is one of them which uses *encryption-based* approach for key exchange. We use the saber key encapsulation mechanism algorithm to propose an improved authentication scheme SKEBA. The phases of the SKEBA are as follows.

A. Set Up Phase

In this phase, the parameters which are used in the Saber key-encapsulation mechanism are computed. This phase is executed by server $\mathcal{S}$ as following.

1) It chooses $q = 2^{\epsilon_q}, p = 2^{\epsilon_p}, R_q = ([\mathbb{Z}_q[X]]/[X^n + 1])$ and $\beta_\chi(R_q^{l \times 1})$.
2) It chooses $seed_A \leftarrow \chi(\{0,1\}^{256})$ and computes $A = gen(seed_A) \in R_q^{l \times l}, s = \beta_\chi(R_q^{l \times 1}), b = ((A^T s + h) \bmod q)$ where a constant vector $h \in R_q^{l \times 1}$ consisting of polynomials all coefficients of which are set to the constant $2^{\epsilon_q - \epsilon_p - 1}$, $pk = (seed_A, b), z = \chi(\{0,1\}^{256}), pkh = H_1(pk)$ and $sk := (z, pkh, pk, s)$.
3) It publishes the list of system parameters $\Delta = \{R_q, p, q, \beta_\chi(R_q^{l \times 1})\}$.

B. Registration Phase

The communication channel utilized during this stage is secure. Upon completion of this phase, a smart card will be issued to the user $\mathcal{U}_i$.

1) $\mathcal{U}_i$ selects two parameters identity ID_i and password PWD_i.
2) It chooses a random integer $r_i \in \mathbb{Z}_q$ and computes $\beta_i = H_1(r_i \| ID_i \| PWD_i)$. Then it sends the values $\{ID_i, \beta_i\}$ to the server $\mathcal{S}$.
3) $\mathcal{S}$ chooses $r_s \in \mathbb{Z}_q$ as a secret value and computes $\gamma_i = H_1(r_s \| ID_i)$, $\lambda_i = \beta_i + \gamma_i$. Then it stores $\{\lambda_i, pk\}$ on a smart card and handed over to the user $\mathcal{U}_i$.
4) When the user $\mathcal{U}_i$ received the smart card, they compute $\gamma_i = \lambda_i \oplus H_1(r_i \| ID_i \| PWD_i)$, $\eta_i = H_3(ID_i \| PWD_i) \oplus r_i$ and $\zeta_i = H_2(H_3(ID_i \| PWD_i) \| r_i \| \gamma_i)$. Then, they store $\{\lambda_i, \eta_i, \zeta_i, pk\}$ on their smart card.

C. Login Phase

1) $\mathcal{U}_i$ inserts their smart card into a reader and inputs their unique ID_i and PWD_i. The smart card calculates $r_i' = H_1(ID_i \| PWD_i) \oplus \eta_i$, $\gamma_i' = \lambda_i \oplus H_1(r_i' \| ID_i \| PWD_i)$ and $\zeta_i' = H_2(H_3(ID_i \| PWD_i) \| r_i' \| \gamma_i')$. Then it checks whether $\zeta_i \stackrel{?}{=} \zeta_i'$ is true or false. In case the statement is determined to be false, the session is terminated by the reader. However, if the statement is true, it verifies the user $\mathcal{U}_i$ and jumps to the next step.

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2) The smart card takes a timestamp T_1 and a random number $r_t \in \mathbb{Z}_q$. Then $\mathcal{U}_i$ uses $m = (ID_i \| T_1 \| r_t)$ as plaintext and encrypts it using public key pk. Then it computes common session key K and sends ciphertext $\{ct\}$ to the server $\mathcal{S}$.
3) $\mathcal{S}$ decrypts ct using private key sk and checks time validity $|T_2 - T_1| \leq \Delta T$. Then it checks ID_i and γ_i. If ID_i is a valid identity, $\mathcal{S}$ authenticates user $\mathcal{U}_i$. Then it encrypts the message $m' = (ID_i \| T_1 \| r_t)$ and checks $ct \stackrel{?}{=} ct'$. If it holds true, $\mathcal{S}$ computes common session key K and sends $\{v = H_1(K \| \gamma_i \| T_1 \| r_t)\}$ to the $\mathcal{U}_i$.
4) $\mathcal{U}_i$ checks $v \stackrel{?}{=} H_1(K \| \gamma_i \| T_1 \| r_t)$. If it holds true, it authenticates server $\mathcal{S}$.

D. Access Control Phase

1) The server $\mathcal{S}$ chooses ID_{i_t}-a temporary identity of user $\mathcal{U}_i$- and computes $\theta = H_2(ID_i \| K)$, $R = (ID_i \| ID_{i_t} \| T_3) \oplus \theta$ and $W = H_1(ID_i \| ID_{i_t} \| T_3)$ where T_3 is current timestamp of $\mathcal{S}$. Then it sends $\{R, W\}$ to $\mathcal{U}_i$ and at the same time, it sends $\{ID_{i_t}, T_3, K\}$ to device $\mathcal{D}_j$ through a secure channel.
2) Upon receiving R, $\mathcal{U}_i$ computes $(ID_i' \| ID_{i_t} \| T_3) = R \oplus H_2(ID_i \| K)$ and checks $W \stackrel{?}{=} H_1(ID_i \| ID_{i_t} \| T_3)$, $|T_4 - T_3| \leq \Delta T$. If $ID_i' = ID_i$ then it computes $\pi = H_1(ID_{i_t} \| T_4)$ and session key $K_t = H_1(ID_{i_t} \| K \| T_3)$ and sends $\{\pi, T_4\}$ to device $\mathcal{D}_j$.
3) The device $\mathcal{D}_j$ checks $|T_5 - T_4| \leq \Delta T$ where T_5 is current timestamp of $\mathcal{D}_j$. Then it checks whether $\pi \stackrel{?}{=} \pi'$. If it is true, it authenticates user $\mathcal{U}_i$ and computes the session key $K_t = H_1(ID_{i_t} \| K \| T_3)$.

VI. Security Analysis of the SKEBA Scheme

In this section, we assess the security of SKEBA scheme using both formal and informal analysis methods. Our evaluation of the SKEBA scheme starts by demonstrating its resistance to attacks proposed in this article. As shown in Table II, the SKEBA scheme exhibits superior security performance than several related authentication schemes. Additionally, we employ both the automated verification tool Scyther and the real or random (ROR) model [24] to formally prove the security of our proposed scheme.

A. Informal Security Analysis

1) *Resistance to Impersonation Attack:* In an impersonation attack scenario, the adversary $\mathcal{A}$ attempts to impersonate a legitimate server by receiving the message $\{ct\}$. However, since the decryption of this message and computation of the session key K requires the server's private key sk, the adversary is unable to do so without obtaining it. Furthermore, impersonating a legal user is impossible without possessing their unique identifier ID_i to generate a valid message $\{ct\}$. Therefore, the SKEBA scheme can effectively resist impersonation attacks.
2) *Resistance to De-Synchronization Attack:* The LAAC scheme is vulnerable to de-synchronization attacks, which occur when the adversary $\mathcal{A}$ modifies the message R without the user $\mathcal{U}_i$ detecting the change. To address this vulnerability, we employ a hash value W to ensure the integrity of the message R.
3) *Resistance to Traceability and Anonymity:* In the SKEBA protocol, all communication messages between server $\mathcal{S}$ and user $\mathcal{U}_i$ are altered in each session as they contain at least one of the randomly generated numbers r_t, unique identifier ID_{i_t} or timestamps T_1, T_3, T_4. This ensures that the SKEBA scheme provides traceability and anonymity.
4) *Resistance to a Man-in-the-Middle Attack:* A man-in-the-middle attack involves an attacker intercepting communications between two parties to either eavesdrop on the conversation or modify it without their knowledge. In the SKEBA scheme, server $\mathcal{S}$ utilizes γ_i to verify the validity of ct after decryption, while $\mathcal{U}_i$ uses W to validate message R. As a result, the SKEBA scheme is designed to be secure against man-in-the-middle attacks.
5) *Resistance to Replay Attack:* In this attack, an adversary first eavesdrops and records packets transmitted between two legitimately communicating end-point and then replays these packets at another time. In the SKEBA protocol, the random numbers r_t, ID_{i_t} along with timestamps T_1, T_3, T_4 are used for each session to prevent the replay attack.
6) *Resistance to Smart Card Stolen Attack:* The information $\{\lambda_i, \eta_i, \zeta_i\}$ stored in the smart card are depend on the identity ID_i and password PWD_i. Even if an adversary $\mathcal{A}$ successfully steals a smart card and extracts its stored information, it would be rendered useless in the absence of the accompanying values of ID_i and PWD_i. As such, any attempt to utilize the information by $\mathcal{A}$ would be futile.

TABLE II
Security Comparison

Protocol	ImA	DeA	RA	TA	MIMA	SCSA
[25]	×	✓	×	✓	✓	✓
[26]	✓	✓	✓	✓	✓	✓
[16]	×	×	✓	✓	✓	×
SKEBA scheme	✓	✓	✓	✓	✓	✓

ImA : impersonation attack
DeA : de-synchronization attack
RA : replay attack
TA : traceability attack
MIMA : man-in-middle attack
SCSA : smart card stolen attack

B. Formal Security Analysis

1) ROR Model: We use the ROR model [24] to prove the semantic security of the SKEBA scheme. In the ROR model, there are various queries that an adversary $\mathcal{A}$ can ask. These queries are as follows, where C^i, S^j, and $U^i = C^i \cup S^j$ are a client instance, a server instance, and a participant instance, respectively.

1) *Execute*(C^i, S^j): Through the use of this query, an adversary $\mathcal{A}$ is able to extract the exchanged messages during a legitimate execution of the protocol between a client instance C^i and server instance S^j.

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2) *Send*(U^i, m): In the event of communication interception, an adversary $\mathcal{A}$ may alter a message and redirect it to its intended recipient. This query generates the response message produced by instance U^i upon receiving message m.
3) *Test*(U^i): Let $b \in \{0, 1\}$ be a bit that has been chosen uniformly at random. Upon requesting this query, if $b = 1$, the output is the actual secret value for instance U^i. However, if $b = 0$, the output is a random vector of the same size as the secret value. If the secret value for U^i is undefined, return the undefined symbol $\perp$.
4) *Corrupt*(U^i): This query simulates the smart card stolen attack in which all secret parameters stored in the smart card are extracted by an adversary $\mathcal{A}$ using physical attacks, such as side-channel analysis.

Denoted as *SUCC*, the event where an adversary correctly guesses the value of bit b. The ROR advantage of an adversary $\mathcal{A}$ to breaking the security of the protocol P is defined as follows:

$$Adv_P^{\text{ROR}}(\mathcal{A}) = |2Pr[SUCC] - 1|.$$

We say a protocol P is secure if $Adv_P^{ROR}(\mathcal{A}) \leq \epsilon$, where ϵ is a small value.

As proved in [23, Th. 6.3], the advantage of an adversary $\mathcal{A}$ to breaking the *Saber Encryption* is negligible, hence $Adv_{Saber.PKE}(\mathcal{A}) \leq \epsilon$

Theorem 1: Let $Adv_{imp}(\mathcal{A})$ be the advantage of an adversary $\mathcal{A}$ to break successfully the semantic security of the SKEBA scheme, then we have

$$Adv_{\text{SKEBA},\mathcal{D}}(\mathcal{A}) \leq 2\left(\frac{n_{H_1}^2}{2 * 2^{256}} + C' \cdot q_n^{s'} + Adv_{Saber.PKE}(\mathcal{A})\right)$$

where n_{H_1} is the number of the hash query to $H_1()$, the output size of $H_1()$ is 256 bits, so its output space size is 2^{256}, q_n is the number of the *Send*() query, $\mathcal{D}$ is the size of password space that follows a Zipf's law [22], [27] in terms of frequency distribution and C', s' are Zipf parameters.

Proof: We use a sequence of games $\{G_i\}_{i=0}^{i=3}$ to prove the theorem. At the beginning of this process, we generate a bit b at uniformly random and conceal it from the adversary $\mathcal{A}$. The goal of an adversary $\mathcal{A}$ is to guess the correct value of bit b using the oracle query *Test*().

Game G_0*:* This game represents the real attack against a protocol P by an adversary $\mathcal{A}$, where the objective is to correctly guess the value of b. Therefore the advantage of the adversary $\mathcal{A}$ is

$$Adv_{SKEBA}(\mathcal{A}) = 2Pr[SUCC_0] - 1. \tag{1}$$

Game G_1*:* In this game, first an adversary $\mathcal{A}$ asks oracle query *Execute*(), and then they try to guess the random bit b by asking the oracle query *Test*(). In the SKEBA scheme, the session key K is related to the parameter κ. An adversary $\mathcal{A}$ requires the secret parameters $r_t, T_1,$ and ID_i to drive the parameter κ. Therefore eavesdropping on the exchanged messages does not increase the probability of winning the game G_1 by an adversary $\mathcal{A}$. As a result, we have

$$Pr[SUCC_0] = Pr[SUCC_1]. \tag{2}$$

Game G_2*:* In this game, we use a Hash, Send, and Execute query to simulate the hash function $H_1()$ and consider its collision property. The input and output of the hash function $H_1()$ are available on the public channel and an adversary $\mathcal{A}$ can intercept and modify it by oracle queries *Execute*() and *Send*(). Then $\mathcal{A}$ asks oracle query *Hash*() to find a hash collision. Based on the birthday paradox, the probability of hash collision of the $H_1()$ is $\leq (n_{H_1}^2/[2 * 2^{256}])$, where n_{H_1} is the number of the hash query to $H_1()$ and 2^{256} is output space size of the $H_1()$. Therefore, we have

$$|Pr[SUCC_2] - Pr[SUCC_1]| \leq \frac{n_{H_1}^2}{2 * 2^{256}}. \tag{3}$$

Game G_3*:* When an adversary $\mathcal{A}$ asks the *Corrupt*() query, they can extract secure information $(pk, \lambda_i, \eta_i, \zeta_i)$ stored on the smart card. To derive secret values r_i, r_s and s from (λ_i, η_i) and pk, an adversary $\mathcal{A}$ have to know the values ID_i, PWD_i. According to [28], the probability of correctly guessing these values can be expressed as $C' \cdot q_n^{s'}$. Besides, as mentioned before, the advantage of an adversary $\mathcal{A}$ to breaking the *Saber Encryption* and deriving the secret value s from pk is $Adv_{Saber.PKE}(\mathcal{A})$. In conclusion, we have

$$|Pr[SUCC_3] - Pr[SUCC_2]| \leq C' \cdot q_n^{s'} + Adv_{Saber.PKE}(\mathcal{A}) \tag{4}$$

where q_n is the number of the *Send*() query.

Let an adversary $\mathcal{A}$ try all ways to break the security of the SKEBA scheme and they do not succeed. Therefore, the last way to attack the scheme is to ask the *Test*() query, so the probability of winning the game G_3 is $Pr[SUCC_3] = (1/2)$.

Using (1) we have

$$\begin{aligned} Adv_{\text{SKEBA},\mathcal{D}}(\mathcal{A}) &= 2|Pr[SUCC_0] - \frac{1}{2}| \\ &= 2|Pr[SUCC_0] - Pr[SUCC_3]|. \end{aligned} \tag{5}$$

We can obtain the following equation by applying the triangular inequality with (5)

$$\begin{aligned} Adv_{\text{SKEBA},\mathcal{D}}(\mathcal{A}) \leq 2(&|Pr[SUCC_1] - Pr[SUCC_2]| \\ &+ |Pr[SUCC_3] - Pr[SUCC_2]|). \end{aligned} \tag{6}$$

Using (3) and (4) we have

$$Adv_{\text{SKEBA},\mathcal{D}}(\mathcal{A}) \leq 2\left(\frac{n_{H_1}^2}{2 * 2^{256}} + C' \cdot q_n^{s'} + Adv_{Saber.PKE}(\mathcal{A})\right). \tag{7}$$

■

2) Scyther Tool: There are some automated security protocol analysis tools, such as Scyther, AVISPA, and Tamarin, to verify the security and correctness of the protocols. The Scyther tool [29] operates under the assumption that cryptographic functions (including hash functions, symmetric algorithms, PUFs, etc.) are perfect, which means that an adversary cannot learn any information from hashed or encrypted data.

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TABLE III
SECURITY ANALYSIS RESULT OF THE SKEBA SCHEME WITH SCYTHER

Claim	Status	Comments
Secret ID_i, ID_{i_t}	Ok	No attacks within bounds
Secret ct, v, R, W, π	Ok	No attacks within bounds
Niagree, Nisynch, Alive, Weakagree	Ok	No attacks within bounds

We describe a security protocol by security protocol description language (SPDL) language as input and take a summary report that states whether any attack pattern is found or not. We define functions, parameters, and events (e.g., sending data, receiving data, etc.) of a security protocol in a role specification. An input SPDL code includes a set of roles, such as the user's role, the server's role, and the patient's role. We refer the interested reader to [29] and [30] for more learning about the Scyther tool and SPDL language. The SPDL code of the SKEBA scheme is available in [31]. As illustrated in Table III, the Scyther tool confirms that the SKEBA scheme is capable of withstanding a variety of passive and active attacks.

3) Security Analysis Against Quantum Attack: Based on the security discussions in [23, Th. 6.5], the *Saber Encryption* is secure against the quantum attack. The server S_j uses the hash function H_1 to compute the message v. We know that the hash functions are quantum-safe, therefore the SKEBA scheme is secure against quantum attacks.

VII. PERFORMANCE ANALYSIS

This section presents a performance evaluation of our proposed scheme with respect to both computation and communication overhead. In this work, we use the open-source implementation of the Saber that was designed by Roy and Basso [32]. We implement the SKEBA scheme in the Vivado2018.3 environment for Zynq UltraScale FPGAs. Key encapsulation mechanism (KEM) Saber uses the hash functions SHA3-256, SHA3-512, and SHAKE-128 so as these cores already are implemented, we can call these hash cores in the SKEBA scheme. This reduces the implementation cost of the SKEBA scheme. The FPGA area usage results for the user side of the SKEBA scheme and the SHA3 alone are shown in Table IV. The approximate running time of cryptographic functions is shown in Table V. It is well established that the performance of algorithms is significantly influenced by the choice of the FPGA family for its implementation, as well as the methodology employed to achieve optimal area/time efficiency. Nevertheless, to compare the performance of the protocols, we can use approximate running time. In Table VI, the computational cost for the user, server, and patient of the SKEBA protocol and that of existing protocols is depicted. The protocols in the table are selected because we want to compare the computational cost of the authentication protocols that use different cryptographic methods, such as elliptic curves, lattice, and symmetric algorithms, with our proposed protocol. Table VII depicts a comparison of the communication overhead of our protocol with some existing protocols. We ignore the communication overhead in the registration phase because data are stored in a smart card and then it is sent to the user $\mathcal{U}_i$ through a secure channel. We assume that the length

TABLE IV
AREA USAGE

Block	CLB LUTs	CLB Registers	BRAM Tile
SKEBA scheme	24184	9776	2
SHA3	5724	3088	0

TABLE V
HARDWARE IMPLEMENTATION RESULTS ON XILINX FPGAS

Architecture	Notation	Time(μs)	Reference
SHA-3	T_h	≈ 0.1	[34]
Elliptic curve point multiplication	T_p	≈ 44	[35]
Saber encapsulation	$T_{e/d}$	≈ 26.5	[32]
Matrix vector multiplication	T_{vm}	≈ 18	[36]
QC-MDPC McEliece	$T_{Enc/Dec}$	$Enc \approx 3.4$, $Dec \approx 23.0$	[37]

TABLE VI
COMPUTATIONAL COST COMPARISON (TOTAL TIME IN μs)

Protocol	[26]	[16]	[12]	[33]	SKEBA
Crypto method	elliptic curve	lattice	lattice	Code-based	lattice
User	$8T_h+$ $3T_p$	$6T_h+$ $2T_{vm}$	$12T_h+$ $4T_{vm}$	$4T_h+$ T_{enc}	$7T_h+$ T_e
Server	$6T_h+$ $6T_p$	$5T_h+$ T_{vm}	$9T_h+$ $3T_{vm}$	$5T_h+$ T_{dec}	$8T_h+$ T_d
Patient	$4T_h + 5T_p$	$4T_h$	-	-	$2T_h$
Total cost	$18T_h+$ $14T_p$	$15T_h+$ $3T_{vm}$	$21T_h+$ $7T_{vm}$	$9T_h + T_{enc}$ $+T_{dec}$	$17T_h+$ $2T_{e/d}$
Total time	≈ 618	≈ 55	≈ 128	≈ 27.3	≈ 55

TABLE VII
COMMUNICATION COST COMPARISON

Protocol	[26]	[16]	[12]	[33]	SKEBA
Login & Verification	552	1300	6344	1400	1440
Access control	-	148	-	-	128
Total cost(Byte)	552	1448	6344	1400	1568

of identity, as well as the outputs of hash functions H_1, and H_3 are each 256 bits. The length of the output of hash function H_2, public key pk, private key sk, and cipher text ct are 512 bits, 672, 832, and 736 bytes, respectively. In the authentication phase, the communicated messages between $\mathcal{U}_i$ and server are public key pk, cipher text ct, and a hash output which is $672 + 736 + 32 = 1440$ byte. Two hash outputs W, π and one 512-bit value R are communicated in the access control phase. As a result, the total communication cost of the access control phase is $2\times256+512 = 1024$ bit. The schemes based on the lattice algorithms have less computational cost than the schemes using elliptic curve algorithms but have more communication overhead than that of ECC-based schemes.

VIII. CONCLUSION

In this article, we have comprehensively analyzed the scheme proposed by Gupta et al. Our findings indicate their scheme is susceptible to various security threats, including impersonation, de-synchronization, and smart card theft

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attacks. To address these vulnerabilities, we use Saber—a finalist lattice-based key encapsulation algorithm in round three of the NIST PQC standardization to propose SKEBA, an improved authentication and access control scheme which is well suited for IoT e-health systems. We implement and test the SKEBA scheme on Zynq UltraScale FPGAs using Verilog language and compare its performance with that of some related existing schemes. In the end, we provide security proof of the SKEBA scheme through formal and informal analysis. In our future work, we aim to further improve the performance of the enhanced scheme by reducing both computational and communication overhead.

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